Design and Fabrication of a High-Speed RC Aircraft: Ratwerks Model M

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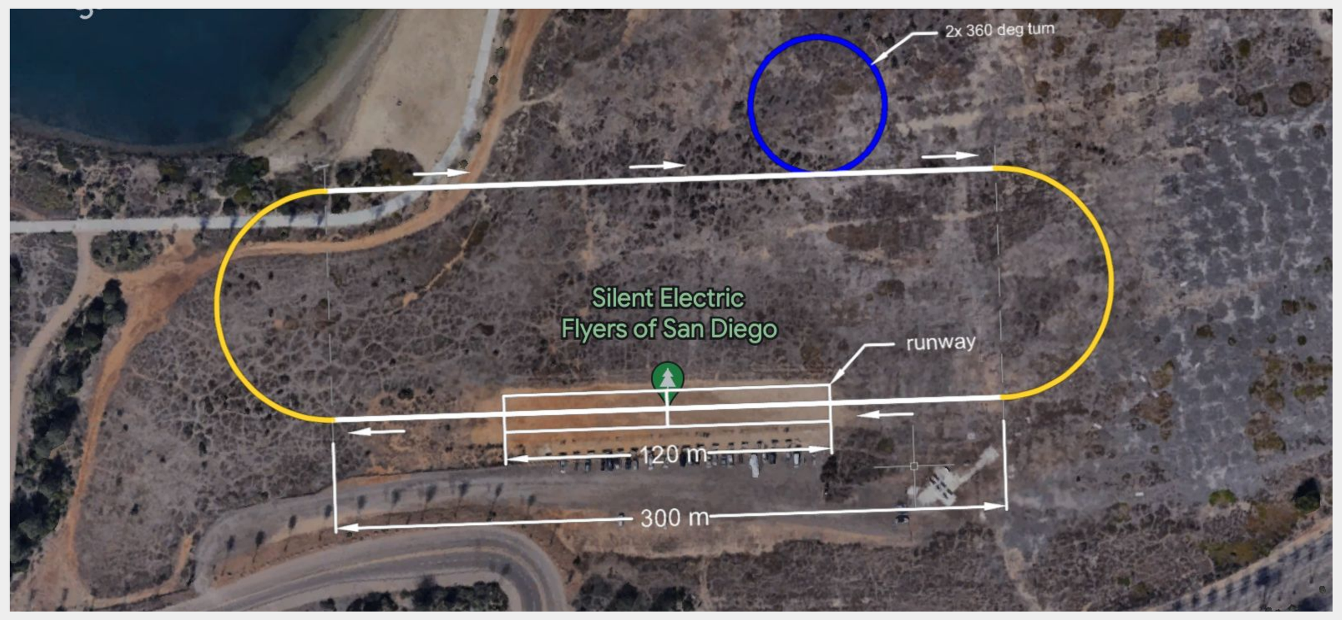
MAE 155B: Aerospace Engineering Design II • Faculty Advisor: Dr. John T. Hwang

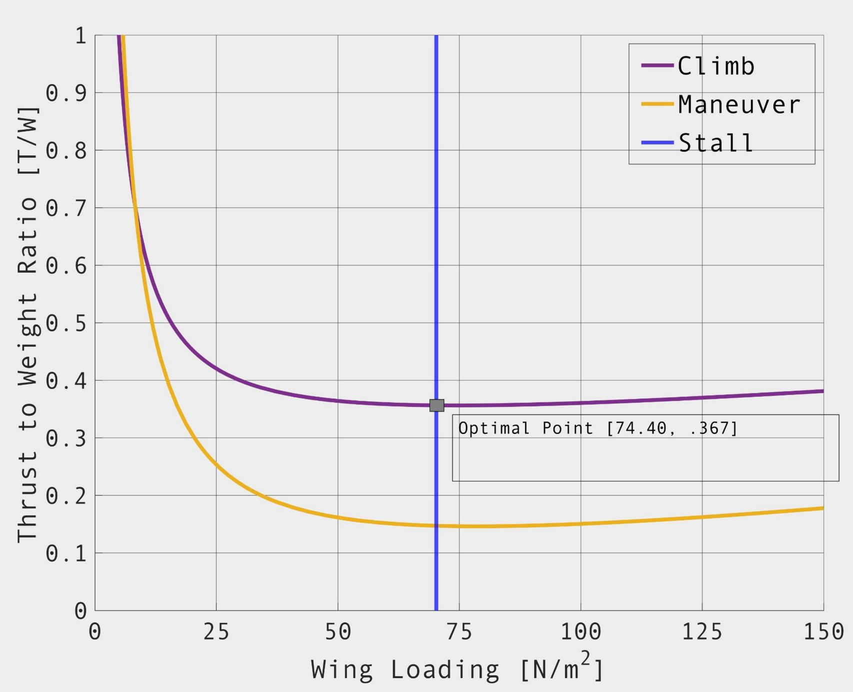
Flight Mission



Location: Silent Electric Flyers of San Diego (Mission Bay)
Mission:

- Takeoff and fly with and without payload according to flight path (above)
- Reach max speed downwind of field
- Perform 2 360° bank turns
- Land safely
- Repeat with payload

Aircraft Sizing



Objectives:

- Maximize W/S (smallest wing) and
- Minimize P/W (smallest motor)
- Satisfy all load constraints

Allows us to get preliminary sizing estimations.

Flight Score

The scoring function is given to judge our flight:

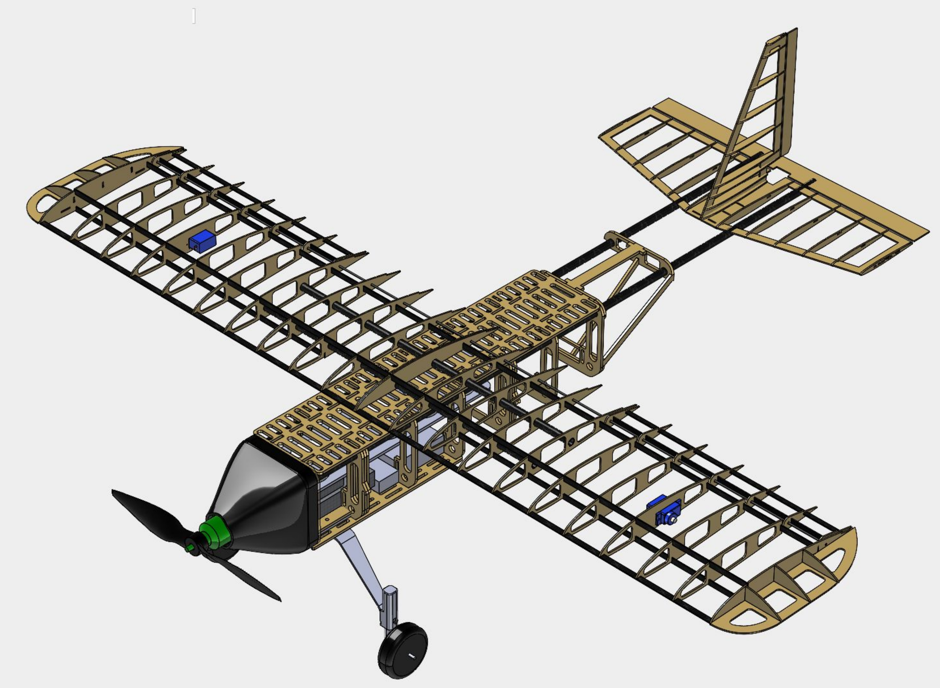
Payload Weight — Max Cruise Velocity

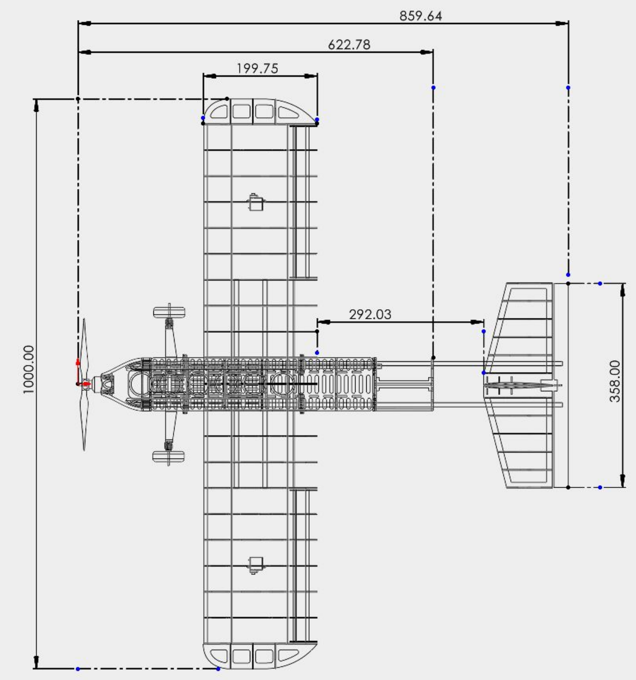
$$Flight\ Score = 7.5\frac{W_p}{W_r} + f_p + \frac{V_{max}}{V_r} + 7.5B$$

Payload Weight Fraction — Empty Flight Bonus (1 or 0)

Variable	Value	Score
Payload Weight (W_p)	2.71 N	20.325
Gross Weight (W_0)	14.22 N	—
Payload Weight Fraction (f_p)	—	.22
Max Cruise Velocity (V_{max})	22.6 m/s	22.6
Empty Flight Bonus (B)	1	7.5
TOTAL SCORE	—	50.6

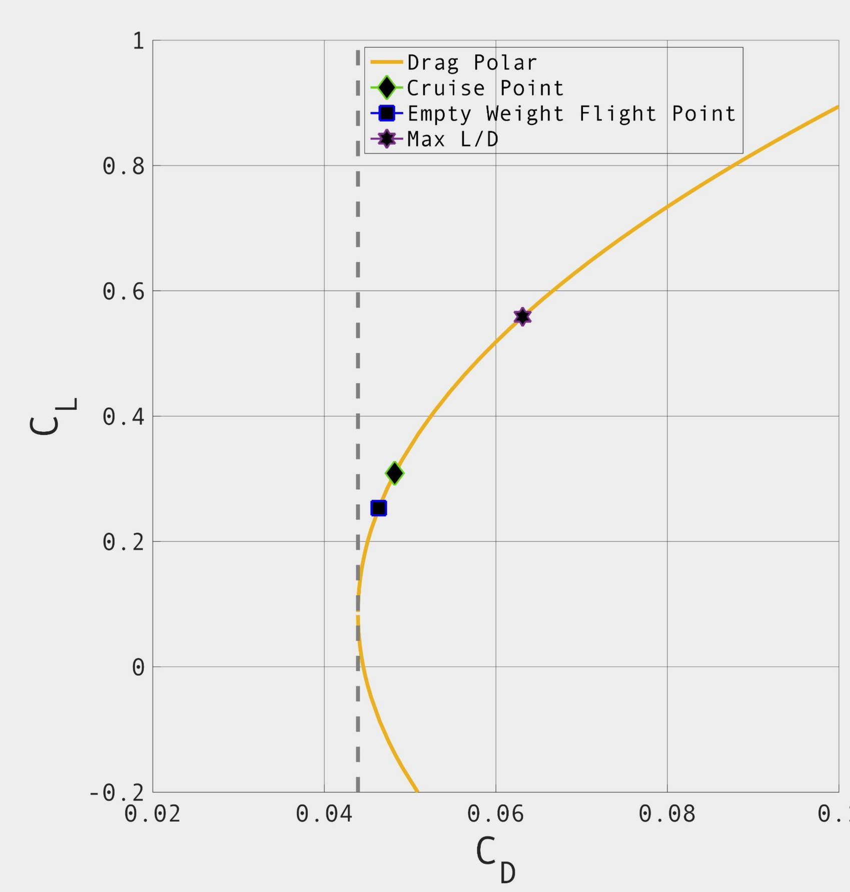
Aircraft Design





	Chord	Span	Area	AR
Wing	20.0 cm	105.5 cm	.2022 m²	5
Horizontal Stabilizer	10.1 cm	36 cm	.0368 m²	3.5
Vertical Stabilizer	10.3 cm	16.9 cm	0.012 m²	2.32
Elevator	6.82 cm	36 cm	.0245 m²	5.3
Rudder	3.5 cm	16.9 cm	0.0051 m²	5.6

Aerodynamics and Propulsion



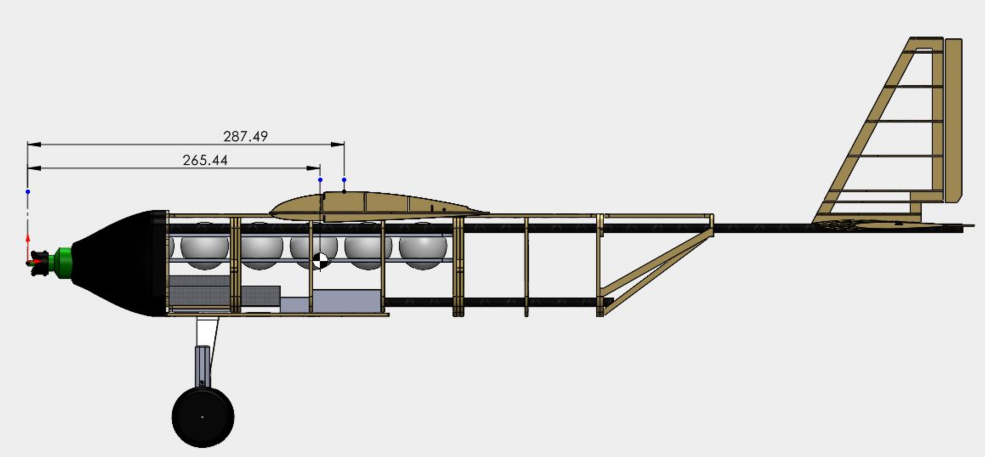
Airfoil Selection: Clark- YM
Thickness-to-Chord: 14%
Max Camber: 3.6%

Parasitic Drag Breakdown	
Wing	0.0082
Horizontal Stabilizer	0.0047
Vertical Stabilizer	0.0025
Fuselage	0.0095
Engine	0.0010
Landing Gear	0.0092
TOTAL	0.0410

Key Aerodynamic Parameters

- C_{Lmax}: 1.326
- $C_{L\alpha}$: 5.133
- **Max Efficiency:** 9.177
- α_{cruise}: ~0°

Weight and Stability

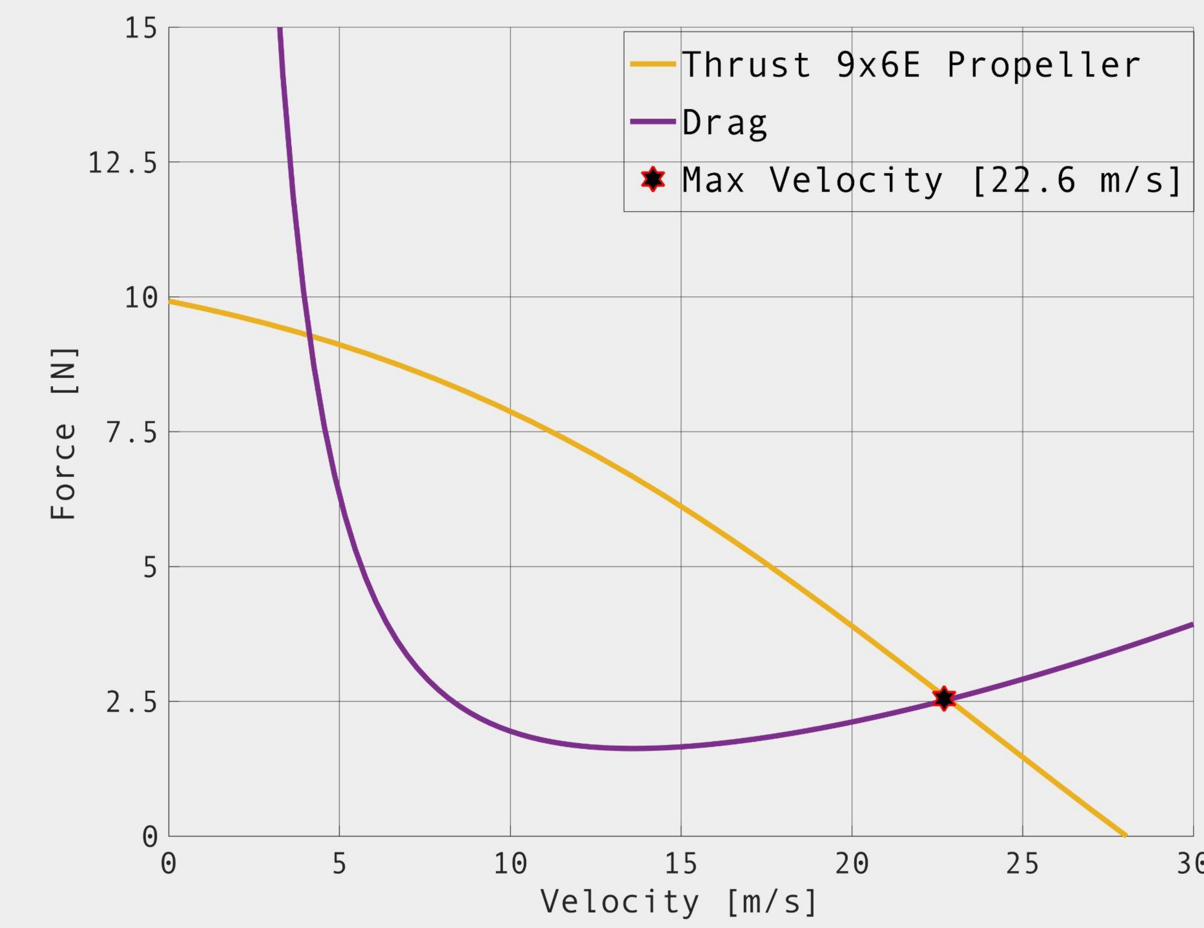


Neutral Point: 287.49 mm

With Payload,
Center of Gravity: 265.44 mm
Static Margin: 13.5%

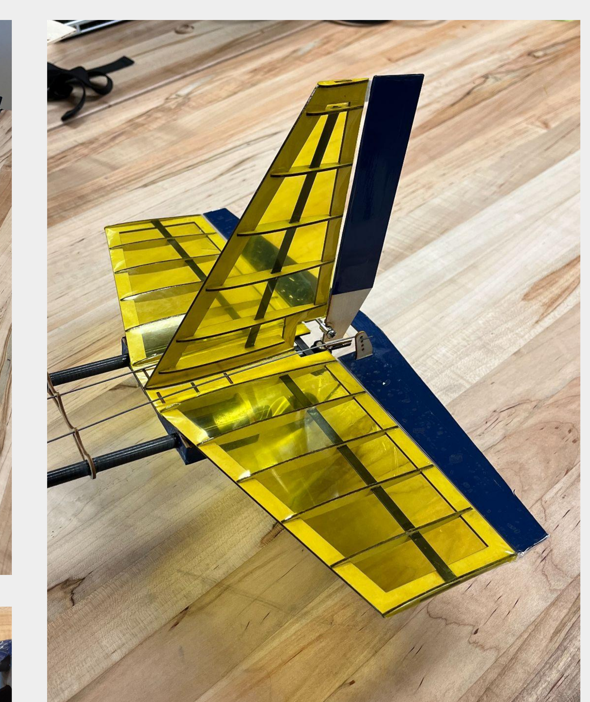
Without Payload,
Center of Gravity: 269.97 mm
Static Margin: 9.3 %

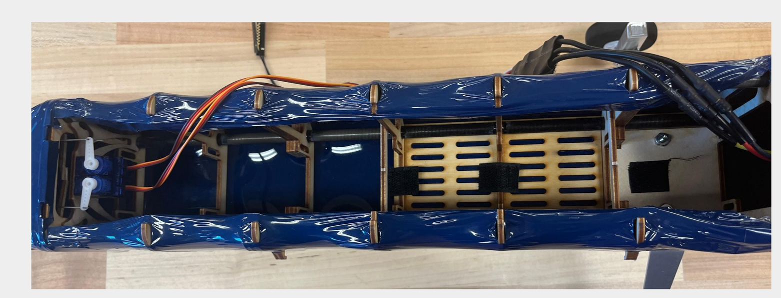
Performance

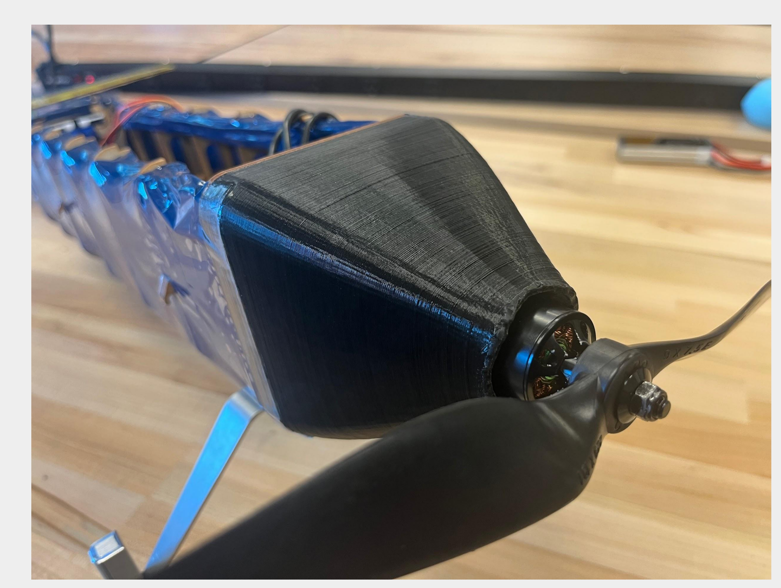


Takeoff Velocity: 10.79 m/s
Takeoff Distance: 15.76 m
Mean Acceleration: 3.7 m/s²
Maximum Velocity: 22.6 m/s

Fabrication







- **Wing, fuselage, and tail surfaces:** Wooden frames reinforced with carbon fiber rods and strips, all wrapped in MonoKote.
- **Interfaces and joints:** Super glue, mechanical locks, notches, bolts, velcro, and rubber bands.
- **Additional features:** Custom-made suspensions on front landing gear to aid during takeoff and landing.